

Sampling:

A warm-up activity

MATH 213

Group 1

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study? **The adults in the United States**
2. What is the sample in this study? **The 765 adults in the sample**
3. What population parameter would someone use this study to estimate? Use the correct notation.
p
4. What sample statistic would we use to estimate this parameter? Use the correct notation.
P-hat 322/765 = 42.09%

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)

Within 5%

Group 2

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study? [Adults in the U.S.](#)
2. What is the sample in this study? [Randomly selected 765 adults in the U.S.](#)
3. What population parameter would someone use this study to estimate? Use the correct notation. [proportion – \$p\$](#)
4. What sample statistic would we use to estimate this parameter? Use the correct notation. [Proportion – \$\hat{p}\$ \(322/765 = 0.421\)](#)

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.) [Within 5%??](#)

Group 3

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?

American adults

2. What is the sample?

765 random Americans

3. What population parameter would someone use this study to estimate? Use the correct notation.
P, all Americans.

What sample statistic would we use to estimate this parameter? Use the correct notation.

Phat - 322/765

4. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)

Group 4

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?

US Adults

2. What is the sample in this study?

765 adults

3. What population parameter would someone use this study to estimate? Use the correct notation.

Let x = could not cover \$400 unexpected expense

P_x

4. What sample statistic would we use to estimate this parameter? Use the correct notation.

$\hat{P}_x = 322/765 = 0.421$

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)

Within 5%

Group 5

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?
United States adults
2. What is the sample in this study?
765 adults in the United States
3. What population parameter would someone use this study to estimate? Use the correct notation.
p
4. What sample statistic would we use to estimate this parameter? Use the correct notation.
P-hat $322/765 = 0.42$

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)

Within 20%

Group 6

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?
 - **All adults in the United States**
2. What is the sample in this study?
 - **765 adults**
3. What population parameter would someone use this study to estimate? Use the correct notation.
 - **P**
4. What sample statistic would we use to estimate this parameter? Use the correct notation.
 - **$P = 322/765 = 0.4209$**

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)
 - **I think this would be within 75-80% accurate for the average adult population**

Group 7

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?
The American People
2. What is the sample in this study?
The 765 adults in the U.S.
3. What population parameter would someone use this study to estimate? Use the correct notation.
 p
4. What sample statistic would we use to estimate this parameter? Use the correct notation.
 $\hat{p}$. 0.42 is the proportion of people who can't pay

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)

This guess would be precise due to the large sample size ($\pm 5-10\%$)

Group 8

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?
Adults in the united states
2. What is the sample in this study?
The 322 adults who could not cover the expense.
3. What population parameter would someone use this study to estimate? Use the correct notation.
Someone could use this study to estimate what proportion of people in the United States cannot afford a \$400 unexpected expense without going into debt.
4. What sample statistic would we use to estimate this parameter? Use the correct notation.
The sample statistic we can use to estimate the parameter would be the 322 people who could not cover the \$400 expense meaning that the proportion of the US population that could not cover the expense would be 0.4209

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)
As a rough guess, roughly within 40 percent.

Group 9

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?

American Adults

2. What is the sample in this study?

765 adults in the united states

3. What population parameter would someone use this study to estimate? Use the correct notation.

P , the percentage of the population who could not cover the \$400

4. What sample statistic would we use to estimate this parameter? Use the correct notation.

$\hat{p} = 322/765 = .4209$ is the proportion of the sample who can't pay

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)

Due to the larger sample size, maybe 10%

Group 10

In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt.

1. What is the population of interest in this study?
2. What is the sample in this study?
3. What population parameter would someone use this study to estimate? Use the correct notation.
4. What sample statistic would we use to estimate this parameter? Use the correct notation.

Of #3 and 4, only one of them has a value you can calculate with the information we have. Calculate that number and add it to the appropriate answer.

5. How precisely would you guess that we should expect this sort of survey to estimate the population parameter? (I am asking for a guess that is a number--"within ... percent"; we will learn how to think systematically about this kind of question very soon.)